

2.54mm apart. Eagle uses a pitch of 0.05 inch or 1.27mm by default in the board layout view. Interestingly, most of the components have pins spaced by multiples of 2.54mm. This makes these boards ideal for use as a former.

Let's go through this step by step. Get a general-purpose PCB like the one shown in Fig. 30. Now let's say you want to drill holes for an 8-pin single-in-line header like the one shown in Fig. 31. Place the copper side of the general-purpose board against the copper side of the PCB and align the two holes you just made with the corresponding holes on the general-purpose PCB (see Fig. 32). It is important to use this PCB orientation because it ensures that the drill bit makes almost perfect holes without damaging the copper pads.

Cut two pieces of a 15mm long and 0.9mm thick wire. If that's not available, use 0.8mm thick tin wire as hookup wire. Push one piece of the wire through the first hole and the other one through the eighth hole. Now squeeze the two PCBs using a small bench vise; preferably one with rubber lips, as shown in Fig. 33.

Now using a mini push hand drill, create holes between the first and eighth pins by placing the drill bit in the six holes on the general-purpose board, one at a time. Release the vice, remove the two pieces of the wire and release the two PCBs. Now place the 8-pin header in the holes just drilled. It should fit perfectly.

To drill holes for a 26-pin dual-in-line header like the one shown in Fig. 35, identify the diagonally opposite pads as shown in Fig. 36. Now drill 1mm holes at the centre of the two pads as precisely as you can. Hold the two PCBs together with their copper sides facing each other. Cut two pieces of 0.8mm tin wire, used as hookup wire, 15mm long. Now push one piece of the wire through the first hole and the other one through the 14th hole,

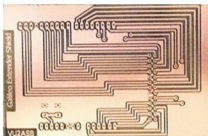


Fig. 29: The PCB

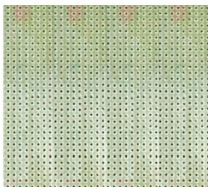


Fig. 30: General-purpose PCB



Fig. 31: 8-pin header block

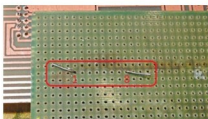


Fig. 32: Aligning the general-purpose PCB with the new PCB



Fig. 33: Squeezing the two PCBs with bench vise

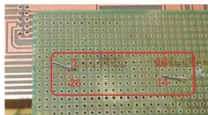


Fig. 34: Aligning for 26-pin DIL header block

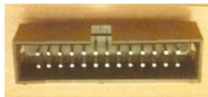


Fig. 35: 26-pin header block

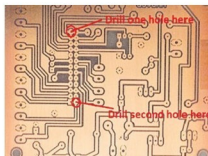


Fig. 36: The diagonally opposite component pads

which is diagonally opposite to the first hole as illustrated in Fig. 34. Now squeeze the two PCBs using a small bench vise; preferably one with rubber lips. Now using a mini push hand-drill drill the balance 24 holes as before.

Release the vice, remove the two pieces of the wire and release the two PCBs. Now place the 26-pin header in the holes just drilled. It should fit perfectly.

Drill all the required holes for the PCB. When done, carefully use very fine sand paper to smoothen the copper side of the PCB. Do not rub too much as the copper pads or tracks may get damaged.

Copper starts oxidising fast. So if you wish to store the PCB for a while, place it in a plastic zipper bag after applying transparent lacquer spray on the copper side. Before use, just wipe the lacquer with nail polish remover. **EFY**